CSM
Fall 2026

EE 64
Worksheet 4

1. Beginning Circuit Analysis

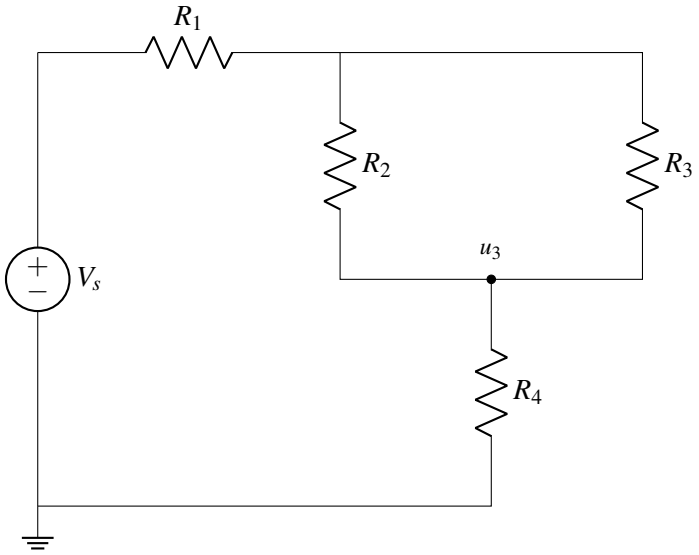
(a) Follow the directions for this subpart to label the circuit according to passive sign convention and NVA.

i. Label each node of the of the following circuit with the following notation of u_i.

ii. Label each side of the circuit components either positive (+) or negative (-).

iii. Represent currents as arrows and draw the currents that flow through each circuit component, denoted as i_i. *Use your labeling from the previous part. Currents flow according to passive sign convention.*

iv. Label the voltages across each circuit, denoted as V_i.



(b) Now that we've correctly labeled our circuit, lets begin the circuit analysis section!

Use KCL to determine the value of u_3 node in terms of any of the following: *other node voltages*, R_1, R_2, R_3, R_4 and V_s (not currents though!!). **Label circuit potential differences and currents.** *Hint: Write out your KCL or KVL equations according to NVA and passive sign convention. How can you manipulate these equations into solving for u_3?*

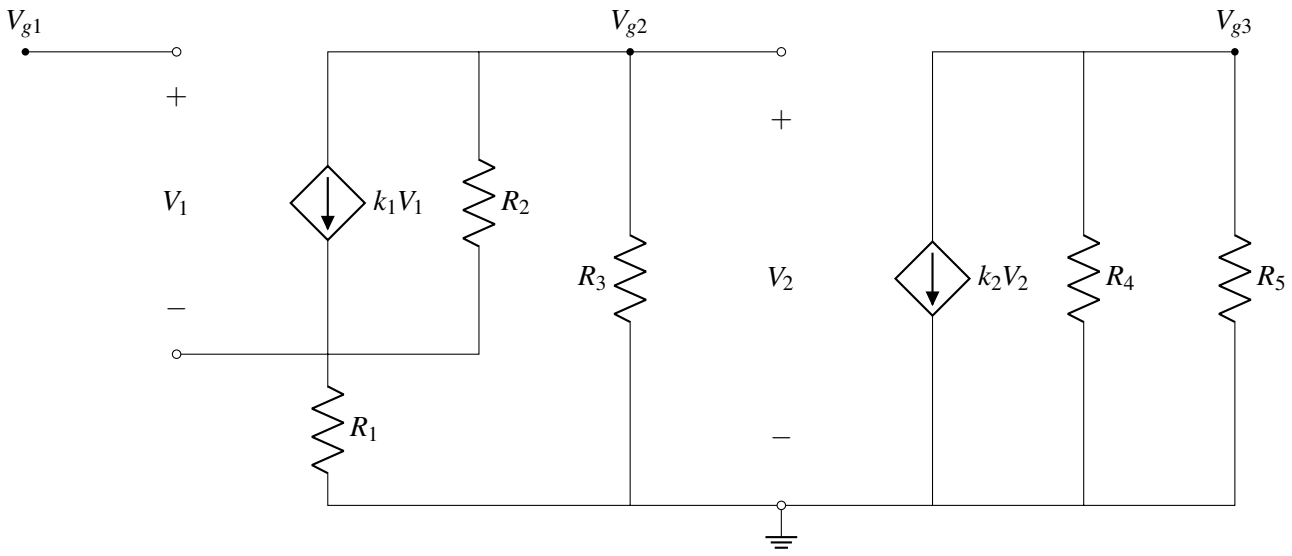
(c) Let's analyze this circuit utilizing KVL now! As a reminder, you can write a KVL equation for a closed loop.

Use KVL to determine the value of u_3 node in terms of *currents (not other node voltages),*, R_1, R_2, R_3, R_4 and V_s. Please do not include node voltages in your final answer.

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2. Kirchhoff's Laws with Dependent Sources

Consider the following circuit:



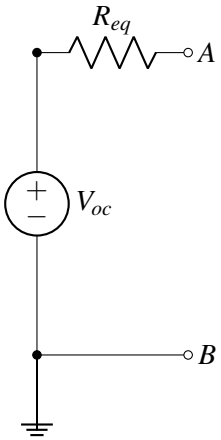
(a) Solve for $\frac{V_{g2}}{V_{g1}}$ using nodal analysis.

(b) Solve for $\frac{V_{g3}}{V_{g2}}$ using nodal analysis.

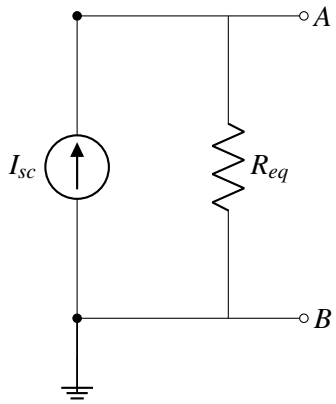
(c) Solve for $\frac{V_{g3}}{V_{g1}}$ using the results above.

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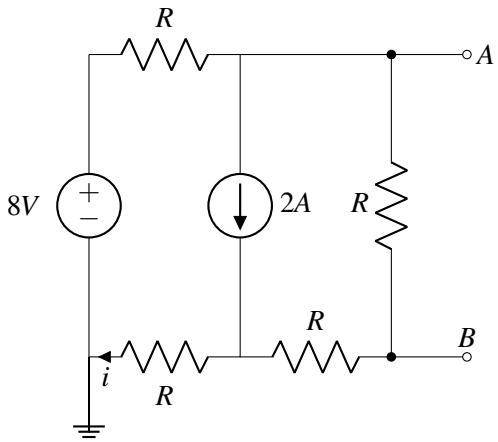
3. Thevenin Norton In this question, you will explore Thevenin Norton equivalence. This is a way of represent the input/output characteristics of a complicated circuit by modeling it as a simple voltage/current source and resistor. Suppose you are given a circuit and 2 terminals A and B. The Thevenin equivalent circuit with respect to those terminals is a circuit of the following form:



where V_{OC} and R_{eq} are set so that the open circuit (OC) voltage between A and B, as well as the short circuit (SC) current through A and B, are the same as in the original circuit. And the Norton equivalent circuit is a circuit of the following form, with the same properties:

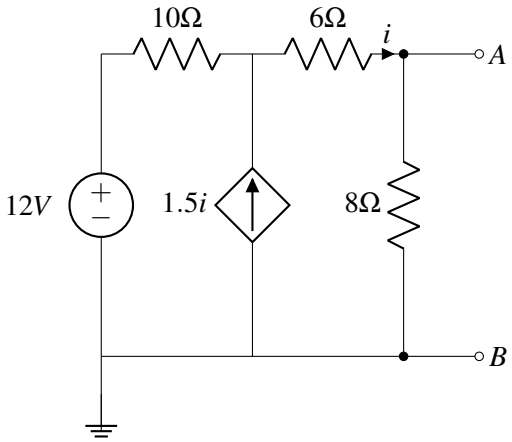


(a) Consider the following circuit. Find the equivalent resistance across the A and B terminal.



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(b) Consider the following circuit. Find the Thevenin and Norton equivalent circuits.



(c) Now, draw the Thevenin and Norton Equivalent Circuit

(d) You can find R_{eq} by taking the ratio of $\frac{V_{OC}}{I_{SC}}$ from the previous part. But in this part, we will find R_{eq} directly by using test voltages and currents. The idea is to zero out all independent sources in the circuit (not dependent sources though). Then, we attach a test voltage V_{test} between A and B and calculate the value of I_{test}, which is the current through your test voltage. R_{eq} will be equal to the ratio $\frac{V_{test}}{I_{test}}$. Find the value of R_{eq} by this method, and compare it with the value of $\frac{V_{OC}}{I_{SC}}$

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